Financial Conditions, r^*, and the Transmission of Monetary Policy Surprises

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Abstract

Interest-rate surprises around Federal Reserve announcements are a clean measure of monetary policy shocks. But their estimated effects on unemployment and inflation since 1994 are often weak or have the wrong sign. We keep the surprise as it is and change only how its effects are traced through the economy. If policy works only through a short-term interest rate, the wrong signs appear. If it works through financial conditions as a whole, measured relative to the neutral interest rate r^*, the effects are standard and precise. A policy move that raises the two-year yield 1 percentage point raises unemployment about 0.4 point and lowers industrial production about 1 percent within a year. It lowers core prices about 0.6 percent after two years. Financial conditions matter most for unemployment, and r^* for prices. The price result relies on data since 2020.

1 Introduction

Measuring monetary policy shocks from market moves around Federal Reserve announcements is a major advance in macroeconomics. Interest-rate futures and Treasury yields change in narrow windows around FOMC announcements. Those changes capture news about policy and little else (Kuttner, 2001; Gürkaynak et al., 2005). Their effects on asset prices are measured precisely (Bernanke and Kuttner, 2005; Hanson and Stein, 2015; Nakamura and Steinsson, 2018; Swanson, 2021).

Using these surprises to measure the effects of policy on output, unemployment and inflation has been harder. In samples that start in the 1990s, the estimated effects are often small, imprecise or of the wrong sign (Ramey, 2016). Recent work has made real progress on this problem. One line of work shows that announcements also reveal the Federal Reserve's view of the economy. A surprise tightening then partly signals good news, and these papers separate the two parts (Romer and Romer, 2000; Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021). Another line shows that part of each surprise can be predicted from public data released before the announcement. Removing that predictable part, and adding surprises from speeches by the Chair, restores standard and larger effects (Bauer and Swanson, 2023a,b). Closest to us, Gertler and Karadi (2015) show that policy surprises work through the cost of credit. Caldara and Herbst (2019) show that leaving out credit spreads, which policy also responds to, weakens the estimated effect on activity. Using futures-based surprises in monthly models goes back to Faust et al. (2004).

This note looks at the other half of the problem: how the effect of a surprise spreads to the economy. We keep the surprise as it is. We ask how much of the post-1994 puzzle comes from two parts of the transmission mechanism that standard models leave out.

First, policy affects spending through financial conditions as a whole: the cost of long-term credit, the value of wealth and the exchange rate (Boivin et al., 2010). It does not work through one short-term rate alone. Policy analysts often summarize these conditions in a financial conditions index (Hatzius et al., 2010; Ajello et al., 2023).

Second, what matters for spending and inflation is the level of real interest rates compared with the neutral rate r^*. The neutral rate is the rate that neither speeds up nor slows down the economy. It fell a lot from the 1990s to the 2010s (Laubach and Williams, 2003; Holston et al., 2017, 2023; Hamilton et al., 2016; Kiley, 2020), and it shaped the long-run trend in yields (Bauer and Rudebusch, 2020). Since 1994, interest rates have trended down along with r^*, while activity has trended up. A simple statistical link between the level of rates and activity then partly picks up these opposite trends. It does not only measure the stance of policy over the business cycle.

The two parts need not matter equally for every outcome. In standard models, activity responds to changes in financial conditions. Inflation depends on how long real rates stay above or below r^*, through the slack this creates. So measuring the stance of policy relative to r^* should matter most for prices.

Our design is simple. The shock is the policy-news surprise of Nakamura and Steinsson (2018), measured in narrow windows around 252 FOMC announcements from 1994 to 2026. We read its effect on the two-year yield, and on a market-based financial conditions index (FCI), from these windows. Unemployment and inflation do not move within the windows. A daily model carries the two-year yield and the FCI forward from the announcement day. Monthly unemployment and core inflation then respond to the monthly averages of these daily series, as in a standard monthly model (a vector autoregression, or VAR).

The model also accounts for when in the month each announcement happens. This matters because a monthly average mixes days before and after the announcement (Kilian, 2024; Chudik and Georgiadis, 2022). Jacobson et al. (2026) show that wrong-signed inflation responses largely disappear when inflation is measured as often as the shock. Related mixed-frequency methods include Francis et al. (2011), Ghysels (2016) and Gazzani and Vicondoa (2020). The FCI and the order of steps—nominal, real, then relative to r^*—come from a companion note on financial conditions (FCI note, 2026). We fixed them before this analysis.

The evidence builds in four steps. Every step includes a smooth time trend (a quadratic) in the monthly equations.

- With the nominal two-year yield as the only channel, the post-1994 puzzles appear. The unemployment response changes with the number of lags and is significantly negative with 12 lags. Prices barely move, and industrial production rises.
- Adding the FCI makes the unemployment response standard, large and significant. But prices fall only a little, and industrial production does not fall.
- Measuring rates in real terms changes little.
- Measuring the two-year yield and the FCI relative to r^* completes the picture. Unemployment rises 0.36 to 0.40 percentage point after 12 months. The core price level falls 0.53 to 0.72 percent after 24 months. Industrial production falls about 1 percent after 12 months. All are significant at the 90 percent level for every number of lags.

We compare these results with the benchmark of Bauer and Swanson (2023a). Our price effect is similar for two years and larger after that. Our effect on industrial production is somewhat smaller. Our effect on unemployment is about twice as large after one year and lasts longer.

The robustness checks show what drives these results. Financial conditions are essential for unemployment. Without the FCI, no adjustment of the two-year yield gives a reliable unemployment

response. With the FCI, every adjustment does. Measuring conditions relative to r^* is essential for prices. The price effect is strong whenever the FCI is measured relative to the Laubach–Williams r^*, however the two-year yield is adjusted. With the nominal or real FCI, the price effect is strong only when the two-year yield is adjusted with a model-based estimate of r^*. The unemployment response holds up however we treat the pandemic, and it is larger in a sample that ends in 2019. The significant price effect, by contrast, depends mainly on data since 2020. The results are the same or slightly stronger with two other surprise measures: the cleaned surprises of Bauer and Swanson (2023a), and the pure-policy surprises of Jarociński and Karadi (2020). They also hold with a fuller method for computing uncertainty.

Our approach complements the work on information effects and predictable surprises. It also shifts what identification rests on. The surprise enters our model only through its immediate effect on the two-year yield and the FCI, and through the timing of the announcement. The rest of the transmission comes from the estimated link between financial conditions and the economy. This makes the results depend less on how strongly the surprise moves rates over a month, and less on its link to earlier news. But it relies on an assumption: the economy responds to a given change in financial conditions in the same way, whatever caused the change. A natural next step is to combine cleaned or separated surprises with our approach. To our knowledge, no earlier study traces the effects of these surprises through a broad financial conditions index measured relative to r^*.

2 Data and method

The surprise. The shock is the Nakamura–Steinsson policy-news factor. It comes from the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025). It summarizes the changes in five short-term interest-rate futures around 252 FOMC announcements from February 1994 to September 2026. The window runs from 10 minutes before the statement to 60 minutes after the press conference starts. When there is no press conference, it is the 30-minute window around the statement. We scale the surprise so that the two-year Treasury yield rises 1 percentage point in the window. We measure the FCI's move in the same windows from the 10-year yield, the S&P 500 and the dollar, using the FCI's weights. The FCI moves 0.99 per percentage point. It is close to one by design, because the weights are scaled so that one unit of the index equals a policy move that raises the two-year yield 1 percentage point. Unemployment and prices do not move in the window.

Financial conditions. The FCI is the market index of FCI note (2026). It is a weighted sum of the 10-year Treasury yield, stock valuations (the cyclically adjusted price-earnings ratio, CAPE) and the real broad dollar. The weights come from how each of these moves with the same policy news. There are three versions, built in the same steps as in that note:

- *Nominal*: the series as measured.
- *Real*: the 10-year yield minus 10-year expected inflation.
- r^**-gap* (baseline): the real 10-year yield minus the Laubach–Williams r^*, and stocks measured by the gap between their earnings yield and r^*. We use the New York Fed's current one-sided estimates of r^*, based on Holston et al. (2023).

The two-year yield is adjusted the same way. It is nominal, then minus two-year expected inflation (from the Cleveland Fed; Haubrich et al., 2012), then also minus r^*. The daily series follow the daily 10-year yield, CAPE and dollar. We shift them smoothly within each month so that their

monthly averages equal the monthly indexes exactly. Expected inflation and r^* move slowly, so the moves in the window are the same in every version.

The model. The model has a daily part and a monthly part. In the daily part, the two-year yield and the FCI (when included) depend on their own averages over the past 1, 5, 22 and 66 days. They also respond to this month's and last month's unemployment and inflation. We estimate this part over 1990–2026. In the monthly part, the unemployment gap and core PCE inflation depend on L lags of themselves and L lags of the monthly averages of the daily series. The unemployment gap is the unemployment rate minus the CBO estimate of its natural rate. We estimate this part by least squares from February 1994 to September 2026, leaving out March–December 2020. We use 3, 6 or 12 monthly lags.

Within each month, unemployment and inflation are set first, and markets then respond to them. In the month of the announcement, unemployment and inflation have not yet moved. To trace the effect of a surprise, we place its window effect on the announcement day. That day is drawn from the actual pattern of FOMC dates within the month. We then run the daily model forward, take monthly averages, and run the monthly model forward. This is a mixed-frequency VAR (Ghysels, 2016) that gives each day in a month the same weight, as a standard monthly VAR does. In large samples, this kind of VAR estimates the same effects as the alternative method of local projections (Plagborg-Møller and Wolf, 2021). A test does not reject equal daily weights for unemployment. It does reject them for inflation, but the gain in fit from unequal weights is very small.

The time trend. Every monthly equation includes a smooth quadratic time trend. Over 1994–2026, unemployment, inflation and interest rates contain slow movements that a short monthly model cannot capture. Rates drifted down along with r^*. Inflation was low in the 2010s and rose after 2020. The natural rate of unemployment may have shifted in ways the CBO estimate misses. Without a trend, these slow movements show up in the estimated effect of the financial variables, and trend gets mixed with transmission. A quadratic is the simplest smooth trend that allows inflation to fall and then rise. Because the trend is an assumption, Appendix Table A3 shows all the main steps without it. The nominal and real steps depend much more on the trend than the r^* step does. This suggests that measuring rates relative to r^* does much of the work of the trend.

The key assumption. We assume the surprise affects unemployment and prices only through the variables in the daily model. We also assume the economy responds to a given change in those variables the same way, whatever caused it. The first part is easier to accept when the financial variables are broad. The second part is the assumption behind any use of a financial conditions index to measure the stance of policy. It is a strong assumption. For example, stock prices partly anticipate future activity. We do not split the FCI into its parts, or into policy-driven and other movements. The index is meant to summarize transmission as a whole, and its weights already measure how each part responds to policy news.

We also tried a method that does not need this assumption: a direct regression of monthly unemployment on the daily surprises (Chudik and Georgiadis, 2022; Chudik and Kilian, 2026). It gives wrong-signed estimates, driven largely by the 2007–09 announcements. It is also about ten times less precise than our estimates, so it cannot tell our estimates apart from zero.

Uncertainty. To measure uncertainty, we repeatedly resample the announcements and the errors of the monthly equations (200 draws). Resampling the announcements also re-estimates the window effects and the timing of announcements. Section 4 reports a fuller version that also re-estimates the daily model and rebuilds the monthly data from the model. The FCI weights and r^* are held fixed throughout. We report 90 percent intervals, as recent studies in this area do (Bauer and Swanson, 2023a; Caldara and Herbst, 2019; Miranda-Agrippino and Ricco, 2021; Ramey, 2016). We also report 68 percent intervals, which Jarociński and Karadi (2020) report

alongside 90 percent ones.

3 Results

Table 1 shows how the unemployment gap and the core PCE price level respond after 12, 24 and 36 months, for 3, 6 and 12 monthly lags, at each step. Figure 1 shows the four key cases.

Table 1: Responses to a policy move that raises the two-year yield 1 percentage point

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	no	0.07	−0.13	−0.27**	0.36†	0.10	−0.19	0.53**	0.29	−0.00
	yes	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
Real	no	0.14†	−0.04	−0.20**	0.33**	0.12	−0.13	0.42**	0.23	−0.01
	yes	0.44**	0.38**	0.29**	0.65**	0.65**	0.59**	0.57**	0.64**	0.64**
r^*-gap	no	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†	0.07
	yes	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
B. Core PCE price level (percent)										
Nominal	no	−0.03	−0.01	−0.02	−0.16	−0.16	−0.07	−0.26†	−0.34	−0.18
	yes	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†	−0.49†
Real	no	−0.14†	−0.08	−0.10	−0.25†	−0.23	−0.16	−0.33†	−0.37†	−0.26
	yes	−0.17†	−0.17†	−0.18†	−0.30†	−0.40†	−0.36†	−0.35†	−0.54†	−0.49†
r^*-gap	no	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
	yes	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**

Columns: months after the policy move, and the number of monthly lags in the model. **: 90 percent interval excludes zero; †: 68 percent interval excludes zero (90 percent does not). Steps are described in Section 2. All versions include a quadratic time trend. Bauer and Swanson (2023a), benchmark (read from their Figure 8): unemployment +0.19, +0.17, +0.15 at 12, 24, 36 months (their 90 percent band excludes zero only at 12 months); CPI (not core PCE) −0.36, −0.48, −0.6 (their bands exclude zero).

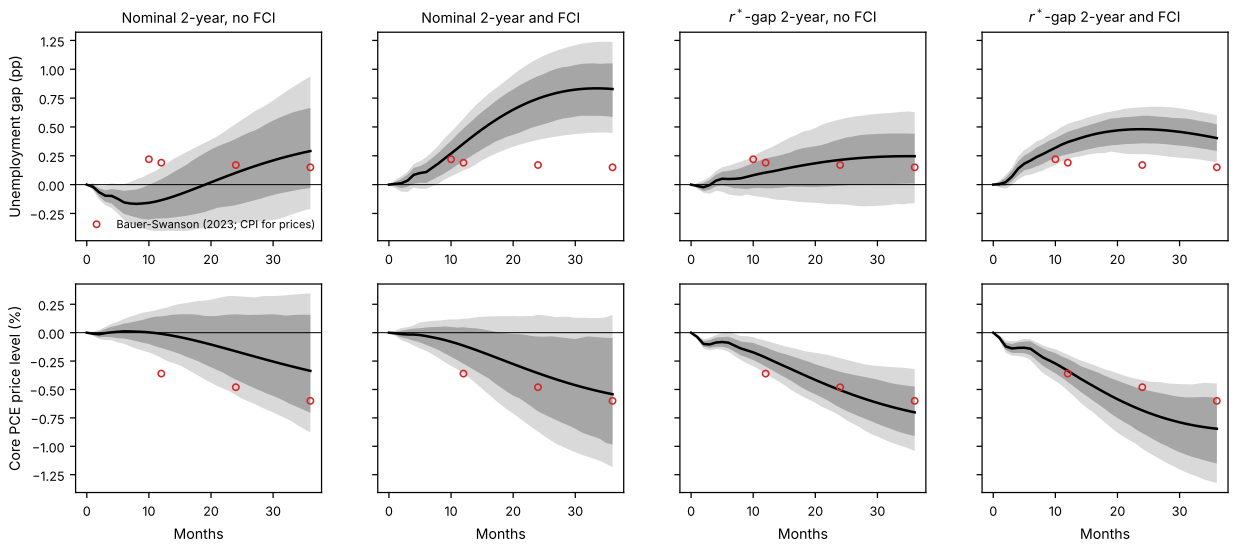


Figure 1: Responses to a policy move that raises the two-year yield 1 percentage point (6 lags). Dark band: 68 percent interval. Light band: 90 percent interval. Circles: Bauer and Swanson (2023a), read from their Figure 8 (unemployment rate; CPI rather than core PCE).

Step 1: the standard approach reproduces the puzzles. Here the nominal two-year yield is the only channel. This is the policy measure that Bauer and Swanson (2023a) use. The unemployment response depends on the number of lags. With 12 lags, it is significantly negative (−0.27 after 12 months), so unemployment falls after a tightening. The price level barely moves. This is the post-1994 pattern documented by Ramey (2016), and the one Bauer and Swanson (2023a) find with surprises they have not cleaned. Industrial production rises significantly (Appendix Table A1).

Step 2: add financial conditions. Adding the FCI makes the unemployment response standard, large and significant at the 90 percent level at every horizon and lag length. Unemployment rises 0.28 to 0.44 percentage point after 12 months and 0.6 to 0.8 after 24 to 36 months. The price level falls 0.2 to 0.5 percent after 24 to 36 months. This fall is significant only at the 68 percent level, and only with 6 and 12 lags. Industrial production does not fall.

Step 3: real rates. Measuring the two-year yield and the FCI net of expected inflation changes little. The companion note finds the same: after 1990, its nominal and real indexes are close. The price fall becomes significant at the 68 percent level at every horizon and lag length.

Step 4: rates relative to r^*. Next we measure the two-year yield and the FCI relative to r^*. Unemployment rises 0.36 to 0.40 percentage point after 12 months, 0.47 to 0.50 after 24 months and 0.35 to 0.43 after 36 months. These numbers are nearly the same for every lag length. The core price level falls 0.33 to 0.34 percent after 12 months, 0.53 to 0.72 percent after 24 months and 0.57 to 0.97 percent after 36 months. All of these are significant at the 90 percent level for every lag length. The unemployment response is smaller and fades sooner than with the nominal FCI. Without the FCI, the r^*-gap two-year yield gives a strong price effect. But the unemployment effect is then weak and depends on the number of lags (it is significant only with 3 lags).

Comparison with Bauer and Swanson (2023a). Their benchmark model covers 1973–2020. It uses surprises from FOMC announcements and Chair speeches, with the predictable part removed. Per percentage point on the two-year yield, it gives an unemployment response of about 0.19 percentage point after 12 months, 0.17 after 24 months and 0.15 after 36 months. It gives a CPI decline of about 0.36, 0.48 and 0.6 percent. Our step-4 price effects (for core PCE) are similar after 12 and 24 months and larger after 36 months (−0.57 to −0.97). Our unemployment effect

is about twice as large after 12 months and 2.5 to 3 times as large after 24 to 36 months. Our intervals exclude zero through 36 months. Theirs include zero after about a year and a half. Their estimates use 30 years of surprises, Chair speeches and removal of the predictable part. Ours use FOMC announcement windows alone, with no such adjustment.

4 Robustness

How should the two-year yield be adjusted? Table 2 and Table 3 keep the FCI as one of the companion note's three indexes. The r^*-gap FCI always uses the Laubach–Williams r^*. The tables change only how the two-year yield is adjusted:

- no adjustment (nominal);
- minus expected inflation and the Laubach–Williams r^* (baseline);
- minus expected inflation and Kiley (2020)'s estimate of r^*;
- minus a market-based measure of the neutral rate: the 9–10-year forward rate from the Federal Reserve's fitted yield curve (Gürkaynak et al., 2007). This rate is nominal, so it replaces both the expected-inflation and r^* adjustments;
- minus the average of the two-year yield over the past 8 years.

Three results stand out.

First, financial conditions are essential for unemployment. Without the FCI, the unemployment response is weak with a model-based r^*. It is significantly negative with the nominal rate, the forward rate or the 8-year average. With any of the FCIs, it is positive at every horizon and significant at the 90 percent level in almost all cases.

Second, the price effect is strong whenever the FCI is measured relative to r^*. With the r^*-gap FCI, the price level falls 0.3 to 0.7 percent after 24 months, however the two-year yield is adjusted, even with no adjustment. This fall is significant at the 90 percent level for every lag length. With the nominal or real FCI, the price effect is strong when the two-year yield is adjusted with a model-based r^*. It is weaker with the forward rate or the 8-year average. The forward rate includes a term premium, which fell over the sample.

Third, Kiley's r^* gives results close to those with Laubach–Williams, though somewhat weaker for prices. Note that the r^*-gap FCI always uses the Laubach–Williams r^*. So these results show that how the two-year yield is adjusted matters less once the FCI is measured relative to r^*. They do not show that the choice of r^* in the FCI is unimportant.

Table 2: Different adjustments of the two-year yield: unemployment gap (percentage points)

	12 months			24 months			36 months		
FCI	3	6	12	3	6	12	3	6	12
2-year: nominal									
none	0.07	−0.13	−0.27**	0.36†	0.10	−0.19	0.53**	0.29	−0.00
nominal	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
real	0.41**	0.32**	0.25†	0.69**	0.67**	0.57**	0.68**	0.74**	0.70**
r^*-gap	0.39**	0.31**	0.29**	0.58**	0.57**	0.55**	0.52**	0.57**	0.60**
2-year: Laubach–Williams r^*-gap (baseline)									
none	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†	0.07
nominal	0.48**	0.43**	0.35**	0.63**	0.63**	0.58**	0.49**	0.53**	0.53**
real	0.46**	0.42**	0.36**	0.59**	0.59**	0.57**	0.45**	0.49**	0.51**
r^*-gap	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
2-year: Kiley r^*-gap									
none	0.13†	−0.02	−0.15†	0.26†	0.07	−0.11	0.30**	0.13	−0.03
nominal	0.44**	0.38**	0.32**	0.58**	0.60**	0.58**	0.43**	0.50**	0.56**
real	0.44**	0.38**	0.32**	0.55**	0.58**	0.57**	0.40**	0.48**	0.54**
r^*-gap	0.39**	0.36**	0.33**	0.45**	0.48**	0.52**	0.31**	0.39**	0.45**
2-year: minus 9–10-year forward rate									
none	−0.20**	−0.20†	−0.23**	−0.25**	−0.17	−0.23†	−0.24**	−0.11	−0.14
nominal	0.26**	0.25**	0.18†	0.33**	0.42**	0.35**	0.26**	0.35**	0.35**
real	0.26**	0.26**	0.20†	0.31**	0.40**	0.35†	0.24**	0.33**	0.34†
r^*-gap	0.22**	0.23**	0.26**	0.26**	0.35**	0.40**	0.21**	0.30**	0.37**
2-year: minus trailing 8-year average									
none	−0.16†	−0.28**	−0.30**	−0.05	−0.18	−0.30†	0.02	−0.06	−0.18
nominal	0.30**	0.25**	0.21**	0.39**	0.43**	0.40**	0.27**	0.35**	0.38**
real	0.27**	0.21**	0.17†	0.36**	0.39**	0.36**	0.26**	0.33**	0.36**
r^*-gap	0.22**	0.17**	0.21**	0.33**	0.36**	0.41**	0.29**	0.37**	0.46**

See Table 1. The FCI is always one of the companion note's three indexes; the r^*-gap FCI uses the Laubach–Williams r^*. All specifications include the quadratic trend.

Table 3: Different adjustments of the two-year yield: core PCE price level (percent)

	12 months			24 months			36 months		
FCI	3	6	12	3	6	12	3	6	12
2-year: nominal									
none	−0.03	−0.01	−0.02	−0.16	−0.16	−0.07	−0.26†	−0.34	−0.18
nominal	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†	−0.49†
real	−0.09	−0.11	−0.13	−0.23	−0.34†	−0.33	−0.31	−0.51†	−0.50
r^*-gap	−0.20**	−0.22**	−0.21**	−0.32**	−0.50**	−0.52**	−0.32†	−0.61†	−0.72**
2-year: Laubach–Williams r^*-gap (baseline)									
none	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
nominal	−0.25**	−0.28**	−0.30**	−0.42**	−0.59**	−0.66**	−0.50**	−0.75**	−0.92**
real	−0.26**	−0.29**	−0.33**	−0.44**	−0.59**	−0.70**	−0.52**	−0.75**	−0.97**
r^*-gap	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**
2-year: Kiley r^*-gap									
none	−0.23**	−0.16†	−0.18**	−0.37**	−0.34†	−0.29†	−0.44**	−0.48†	−0.41†
nominal	−0.23**	−0.24**	−0.21†	−0.39**	−0.50**	−0.41†	−0.43**	−0.61**	−0.53†
real	−0.25**	−0.25**	−0.25**	−0.40**	−0.51**	−0.46†	−0.44**	−0.62**	−0.58†
r^*-gap	−0.34**	−0.33**	−0.29**	−0.48**	−0.62**	−0.58**	−0.47**	−0.69**	−0.68**
2-year: minus 9–10-year forward rate									
none	0.01	−0.04	−0.08	0.04	−0.04	−0.06	0.05	−0.06	−0.07
nominal	−0.05	−0.12	−0.14†	−0.07	−0.21	−0.25	−0.06	−0.22	−0.31
real	−0.06	−0.14†	−0.18†	−0.07	−0.21	−0.30†	−0.06	−0.21	−0.37
r^*-gap	−0.22**	−0.27**	−0.30**	−0.29**	−0.48**	−0.62**	−0.30†	−0.53**	−0.81**
2-year: minus trailing 8-year average									
none	−0.08	−0.06	−0.08	−0.25†	−0.27†	−0.23†	−0.37†	−0.47†	−0.45†
nominal	−0.13†	−0.15†	−0.16†	−0.27†	−0.38**	−0.39†	−0.31†	−0.49**	−0.56†
real	−0.14†	−0.15†	−0.18†	−0.28†	−0.37**	−0.42†	−0.32†	−0.49**	−0.61†
r^*-gap	−0.25**	−0.24**	−0.24**	−0.46**	−0.61**	−0.63**	−0.56**	−0.83**	−0.96**

See Table 2. The 8-year average and the Laubach–Williams trend in the two-year yield (two-year expected inflation plus r^*) are close until 2015 (averages 5.3 and 5.5 percent over 1994–2007; 2.3 and 2.5 over 2008–2015) and diverge after 2016 (1.4 and 3.6).

The pandemic and recent years. The baseline leaves out March–December 2020 as outcomes. But 2020 values still enter the 2021 equations as lags. Table 4 therefore also drops 2021. It then ends the sample in December 2022, and separately in December 2019. In those two shorter samples, the daily model is also estimated only over the shorter period.

With the r^*-gap two-year yield and FCI, the unemployment response holds up. After 12 months, it is 0.26 to 0.31 percentage point when 2021 is dropped, 0.32 to 0.34 when the sample ends in 2022, and 0.42 to 0.44 when it ends in 2019. It is significant at the 90 percent level at almost every horizon and lag length. Without the FCI, it is insignificant in all three samples.

The price effect is unchanged when the sample ends in 2022, so the data since 2023 play no special role. It stays significant with 6 and 12 lags when 2021 is dropped (−0.43 and −0.63 percent after 24 months). But in the sample that ends in 2019, the price level barely moves. So the significant price effect comes from including 2020–22.

Figure 2 shows why these years are informative. After 2021, the r^*-gap two-year yield and the FCI rose by about 4 points. This was the largest swing in the stance of policy relative to r^* in the sample, and core inflation then fell. Figure 3 drops one calendar year at a time. No single year drives the results. The unemployment response after 12 months ranges from 0.26 (dropping

2021) to 0.41. The price level after 24 months ranges from −0.43 (dropping 2021) to −0.86. The 90 percent intervals exclude zero in every case. So the price result rests on the 2020–22 episode as a whole—a large tightening relative to r^* followed by falling inflation—not on any one year.

Table 4: Different samples: r^*-gap two-year yield, without and with the r^*-gap FCI

		12 months			24 months			36 months		
Sample	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Baseline	none	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†	0.07
	r^*-gap	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
Also drop 2021	none	0.03	−0.03	−0.09	0.09	−0.01	−0.09	0.12	−0.01	−0.11
	r^*-gap	0.31**	0.26**	0.30**	0.38**	0.33**	0.38**	0.29**	0.28†	0.30**
End in 2022	none	0.11	0.07	−0.04	0.19†	0.11	−0.01	0.21†	0.13	0.01
	r^*-gap	0.32**	0.33**	0.34**	0.35**	0.38**	0.43**	0.23**	0.30**	0.37**
End in 2019	none	0.05	0.05	0.00	0.09	0.14	0.14	0.11	0.19	0.28
	r^*-gap	0.42**	0.42**	0.44**	0.45**	0.50**	0.60**	0.34**	0.42**	0.59**
B. Core PCE price level (percent)										
Baseline	none	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
	r^*-gap	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**
Also drop 2021	none	−0.11†	−0.09	−0.13†	−0.18†	−0.25†	−0.35**	−0.22†	−0.38†	−0.59**
	r^*-gap	−0.16†	−0.21**	−0.28**	−0.24†	−0.43**	−0.63**	−0.29†	−0.60**	−0.93**
End in 2022	none	−0.25**	−0.20**	−0.21**	−0.42**	−0.49**	−0.51**	−0.51**	−0.70**	−0.83**
	r^*-gap	−0.34**	−0.33**	−0.30**	−0.51**	−0.68**	−0.71**	−0.56**	−0.87**	−1.02**
End in 2019	none	0.02	0.02	0.08†	0.06	0.02	0.03	0.10	0.02	−0.06
	r^*-gap	−0.02	−0.03	−0.05	0.02	−0.00	−0.12	0.03	−0.03	−0.22†

See Table 1. In the "End in" samples, the daily model is also estimated over the shorter period.

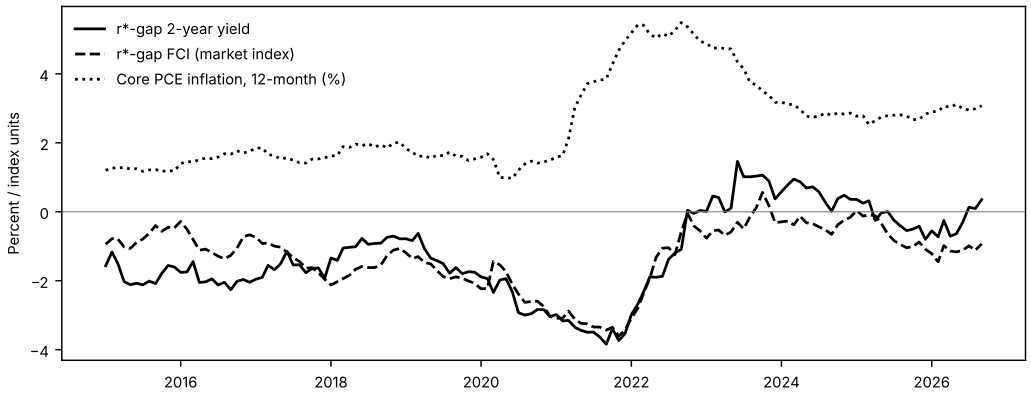


Figure 2: The r^*-gap two-year yield, the r^*-gap FCI and 12-month core PCE inflation, 2015–2026.

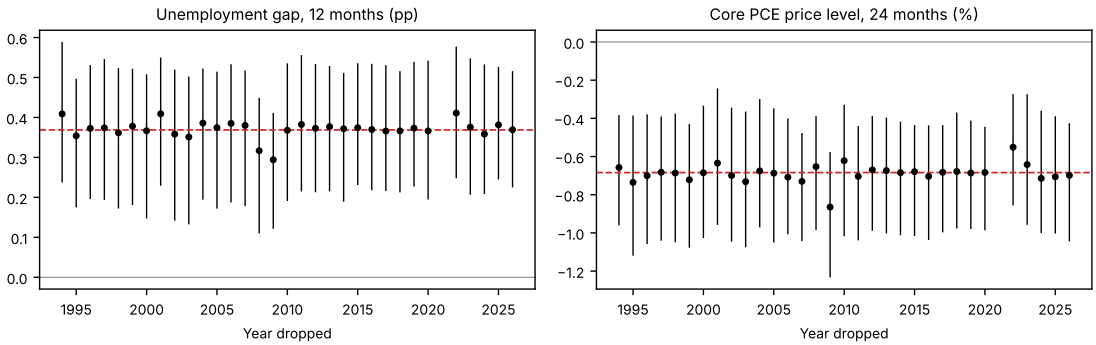


Figure 3: Dropping one year at a time. Baseline version (r^*-gap two-year yield and FCI, 6 lags, 100 draws). Each point drops one calendar year from the monthly equations. Bars: 90 percent intervals. Dashed line: full-sample estimate.

Industrial production. Appendix Table A1 adds industrial production to the monthly model. Without the FCI, industrial production rises after a tightening with the nominal and real two-year yield, and does not move significantly with the r^*-gap two-year yield. With the nominal or real FCI, it does not fall significantly. Only at the r^* step does it fall: by 0.84 to 1.05 percent after 12 months, significant at the 90 percent level for every lag length. Bauer and Swanson (2023a) find about 1.5 percent. In our model, industrial production recovers within about three years. In earlier exploratory work, we also added a commodity price index. Commodity prices then rose after a tightening. Our key assumption is least plausible for prices like these, which look ahead and respond strongly to demand.

Other surprise measures. Appendix Table A2 replaces the Nakamura–Steinsson surprise with three others and keeps everything else the same:

- the surprise with the part predicted by the Fed staff's Greenbook forecasts removed;
- the surprise of Bauer and Swanson (2023a), with the part predicted by public data removed (their posted series, which covers 201 of our announcements through 2019);
- the "pure policy" surprise of Jarociński and Karadi (2020), which keeps only announcements where the two-year yield and the S&P 500 move in opposite directions (157 announcements).

The choice of surprise series can matter for estimated effects (Bu et al., 2021; Brennan et al., 2025). In our design, a surprise enters only through its window moves, scaled to a 1 point rise in the two-year yield, and through the timing of announcements. So this check really tests how much the FCI moves per point on the two-year yield. That ratio is 0.99 for the baseline, 1.07 for the Bauer–Swanson surprise and 1.09 for the Jarociński–Karadi surprise. The last ratio is higher partly by design: that surprise keeps announcements where stocks fall when yields rise, and stocks enter the FCI with a negative sign.

With these ratios, the effects are slightly larger than in the baseline. At the r^* step, unemployment rises 0.40 to 0.45 percentage point after 12 months, and the price level falls 0.55 to 0.77 percent after 24 months. The Greenbook-cleaned surprise gives results within 0.02 of the baseline. Without the FCI, all the surprises give almost identical effects. So our results do not depend on how the surprise is cleaned of information or predictable parts, which matter here only through the ratio of the window moves.

A fuller measure of uncertainty. Appendix Table A4 uses 1,000 draws instead of 200. In each draw, besides resampling the announcements, it re-estimates the daily model on resampled

blocks of 66 days. It also rebuilds the monthly data from the estimated model, randomly flipping the signs of the errors, and re-estimates the monthly equations. The intervals barely change. At the r^* step, every significance marker is the same. Elsewhere, a few markers move between the 68 and 90 percent levels. The FCI weights and r^* are still held fixed.

Other checks. In earlier versions of the model, we added the gap between Baa corporate and Treasury yields to the daily model. This lowered the unemployment response by a third to a half. But no intraday credit data exist, so we could not measure how the credit spread moves in the announcement window. We do not pursue this here. Credit is the most important channel we leave out (Gilchrist and Zakrajšek, 2012; Caldara and Herbst, 2019). We also tried replacing the daily model's path for the financial variables in the first one to two months with the path estimated directly from the data. That gives larger but much less precise effects.

5 Conclusion

Interest-rate surprises around Federal Reserve announcements measure policy shocks well. The hard part has been tracing their effects on the economy. We keep the surprise as it is and change only how its effects are traced. Two standard parts of the transmission mechanism explain much of the post-1994 puzzle. Policy works through financial conditions as a whole. And what matters for spending and inflation is the level of rates and financial conditions relative to a slow-moving neutral rate. When we trace surprises through a broad financial conditions index measured relative to r^*, the effects on unemployment, prices and industrial production have the standard signs. They are precise, they barely depend on the number of lags, and they are similar in size to the best recent estimates.

The two parts play different roles. Financial conditions are essential for the unemployment effect, in every version and sample we try. Measuring financial conditions relative to r^* is essential for the price effect. When the FCI is measured relative to the Laubach–Williams r^*, the price effect is strong however the two-year yield is adjusted. The unemployment effect holds up however we treat the pandemic. The price effect depends mainly on the years since 2020, when the stance of policy relative to r^* moved most.

These findings complement the work on information effects and predictable surprises. Several mechanisms can operate at once. A natural next step is to trace cleaned or separated surprises through our approach. Our results have limits. They rest on the key assumption: policy reaches the economy through the financial variables in the model, and the economy responds to them the same way whatever moved them. They also rest on the time trend. They leave out credit spreads. And our intervals do not include uncertainty about r^* or the FCI weights. The broader lesson is that the fall in r^* over the past three decades is more than a fact about the level of interest rates. It also makes the effects of policy harder to measure, unless rates are measured relative to r^*.

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A Additional tables

Table A1: Monthly model with industrial production added

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	no	0.07	−0.04	−0.19†	0.26†	0.16	−0.04	0.26	0.30†	0.19
	yes	0.41**	0.36**	0.22**	0.60**	0.59**	0.47**	0.51**	0.56**	0.49**
Real	no	0.12	0.03	−0.14†	0.24†	0.15	−0.02	0.24†	0.23	0.15
	yes	0.41**	0.37**	0.24**	0.54**	0.53**	0.46**	0.43**	0.48**	0.45**
r^*-gap	no	0.16†	0.16†	0.03	0.30**	0.27†	0.14	0.33**	0.30†	0.25†
	yes	0.44**	0.42**	0.34**	0.52**	0.50**	0.50**	0.42**	0.42**	0.42**
B. Industrial production (percent)										
Nominal	no	1.06**	1.26**	1.65**	1.03†	1.26†	1.64**	1.16†	1.00†	1.10†
	yes	−0.16	−0.29	0.11	−0.09	−0.04	0.15	0.45	0.39	0.43
Real	no	0.64†	0.77**	1.13**	0.72	0.97†	1.28**	0.87†	0.85†	0.91†
	yes	−0.33	−0.52	−0.23	−0.11	−0.11	−0.01	0.42	0.31	0.34
r^*-gap	no	0.13	−0.04	0.27	0.13	0.23	0.32	0.31	0.44	0.21
	yes	−0.84**	−1.05**	−0.95**	−0.46†	−0.48	−0.80**	0.18	0.08	−0.21

See Table 1. Price-level responses are similar to Table 1. Bauer and Swanson (2023a), industrial production: about −1.5, −1.0 and −0.85 at 12, 24 and 36 months, read from their Figure 8.

Table A2: Other surprise measures, with the FCI

		12 months			24 months			36 months		
Shock	Step	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nakamura–Steinsson	nominal	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
	r^*-gap	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
Greenbook-purged NS	nominal	0.43**	0.34**	0.27**	0.75**	0.73**	0.62**	0.75**	0.81**	0.76**
	r^*-gap	0.39**	0.36**	0.34**	0.46**	0.47**	0.48**	0.34**	0.39**	0.42**
Bauer–Swanson orth.	nominal	0.47**	0.39**	0.32**	0.81**	0.79**	0.67**	0.81**	0.87**	0.80**
	r^*-gap	0.44**	0.41**	0.40**	0.52**	0.52**	0.54**	0.40**	0.44**	0.46**
Jarociński–Karadi	nominal	0.48**	0.40**	0.33**	0.82**	0.81**	0.69**	0.83**	0.89**	0.81**
	r^*-gap	0.45**	0.42**	0.41**	0.53**	0.54**	0.55**	0.41**	0.45**	0.47**
B. Core PCE price level (percent)										
Nakamura–Steinsson	nominal	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†	−0.49†
	r^*-gap	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**
Greenbook-purged NS	nominal	−0.08	−0.11	−0.11	−0.24	−0.35†	−0.30	−0.33	−0.54†	−0.48
	r^*-gap	−0.34**	−0.33**	−0.33**	−0.52**	−0.67**	−0.71**	−0.56**	−0.83**	−0.95**
Bauer–Swanson orth.	nominal	−0.09	−0.12	−0.12	−0.25	−0.37†	−0.32	−0.34	−0.55†	−0.51
	r^*-gap	−0.36**	−0.35**	−0.35**	−0.55**	−0.71**	−0.76**	−0.59**	−0.88**	−1.01**
Jarociński–Karadi	nominal	−0.09	−0.12	−0.12	−0.25	−0.37†	−0.33	−0.34	−0.56†	−0.51
	r^*-gap	−0.36**	−0.36**	−0.35**	−0.55**	−0.72**	−0.77**	−0.60**	−0.89**	−1.03**

See Table 1. Each surprise is scaled to raise the two-year yield 1 percentage point in the window. Step: the nominal, or the r^*-gap, two-year yield and FCI.

Table A3: Main steps without the time trend

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	no	$-0.13^{\dagger}$	-0.27^{**}	-0.34^{**}	-0.01	$-0.19^{\dagger}$	-0.36^{**}	0.08	-0.08	$-0.25^{\dagger}$
	yes	0.18^{**}	0.09	0.07	0.38^{**}	0.29^{**}	$0.19^{\dagger}$	0.42^{**}	0.37^{**}	0.24^{**}
Real	no	-0.02	$-0.14^{\dagger}$	-0.25^{**}	0.07	-0.10	-0.28^{**}	0.12	-0.03	$-0.21^{\dagger}$
	yes	0.25^{**}	$0.18^{\dagger}$	0.12	0.40^{**}	0.33^{**}	0.24^{**}	0.38^{**}	0.36^{**}	0.26^{**}
r^*-gap	no	0.08	0.03	-0.08	$0.16^{\dagger}$	0.04	$-0.15^{\dagger}$	$0.20^{\dagger}$	0.05	-0.16
	yes	0.30^{**}	0.32^{**}	0.34^{**}	0.38^{**}	0.39^{**}	0.41^{**}	0.34^{**}	0.38^{**}	0.37^{**}
B. Core PCE price level (percent)										
Nominal	no	0.01	0.06	0.01	-0.08	0.03	0.05	-0.17	-0.07	0.03
	yes	-0.09	-0.07	-0.10	$-0.28^{\dagger}$	$-0.24^{\dagger}$	-0.23	$-0.42^{\dagger}$	$-0.41^{\dagger}$	$-0.36^{\dagger}$
Real	no	$-0.10^{\dagger}$	-0.02	-0.07	$-0.19^{\dagger}$	-0.08	-0.06	$-0.26^{\dagger}$	-0.16	-0.08
	yes	$-0.17^{\dagger}$	$-0.14^{\dagger}$	-0.18^{**}	$-0.33^{\dagger}$	$-0.32^{\dagger}$	$-0.31^{\dagger}$	$-0.44^{\dagger}$	-0.47^{**}	$-0.41^{\dagger}$
r^*-gap	no	$-0.15^{\dagger}$	$-0.13^{\dagger}$	-0.19^{**}	$-0.22^{\dagger}$	$-0.26^{\dagger}$	$-0.31^{\dagger}$	$-0.28^{\dagger}$	-0.39	$-0.44^{\dagger}$
	yes	$-0.18^{\dagger}$	-0.25^{**}	-0.33^{**}	$-0.28^{\dagger}$	-0.48^{**}	-0.61^{**}	$-0.37^{\dagger}$	-0.68^{**}	-0.84^{**}

As Table 1, but without the quadratic time trend.

Table A4: Main steps with a fuller measure of uncertainty

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	no	0.07	$-0.13^{\dagger}$	-0.27^{**}	$0.36^{\dagger}$	0.10	$-0.19^{\dagger}$	0.53^{**}	0.29	-0.00
	yes	0.44^{**}	0.36^{**}	0.28^{**}	0.76^{**}	0.75^{**}	0.63^{**}	0.76^{**}	0.83^{**}	0.77^{**}
Real	no	$0.14^{\dagger}$	-0.04	-0.20^{**}	0.33^{**}	0.12	$-0.13^{\dagger}$	0.42^{**}	0.23	-0.01
	yes	0.44^{**}	0.38^{**}	0.29^{**}	0.65^{**}	0.65^{**}	0.59^{**}	0.57^{**}	0.64^{**}	0.64^{**}
r^*-gap	no	0.19^{**}	0.10	-0.03	0.34^{**}	$0.21^{\dagger}$	0.04	0.38^{**}	$0.25^{\dagger}$	0.07
	yes	0.40^{**}	0.37^{**}	0.36^{**}	0.47^{**}	0.48^{**}	0.50^{**}	0.35^{**}	0.40^{**}	0.43^{**}
B. Core PCE price level (percent)										
Nominal	no	-0.03	-0.01	-0.02	-0.16	-0.16	-0.07	-0.26	-0.34	-0.18
	yes	-0.09	-0.11	-0.11	-0.24	$-0.36^{\dagger}$	$-0.31^{\dagger}$	-0.33	$-0.54^{\dagger}$	$-0.49^{\dagger}$
Real	no	$-0.14^{\dagger}$	-0.08	-0.10	$-0.25^{\dagger}$	$-0.23^{\dagger}$	-0.16	$-0.33^{\dagger}$	$-0.37^{\dagger}$	-0.26
	yes	$-0.17^{\dagger}$	$-0.17^{\dagger}$	$-0.18^{\dagger}$	$-0.30^{\dagger}$	$-0.40^{\dagger}$	$-0.36^{\dagger}$	$-0.35^{\dagger}$	$-0.54^{\dagger}$	$-0.49^{\dagger}$
r^*-gap	no	-0.27^{**}	-0.22^{**}	-0.23^{**}	-0.45^{**}	-0.51^{**}	-0.52^{**}	-0.55^{**}	-0.70^{**}	-0.80^{**}
	yes	-0.34^{**}	-0.33^{**}	-0.33^{**}	-0.53^{**}	-0.68^{**}	-0.72^{**}	-0.57^{**}	-0.85^{**}	-0.97^{**}

As Table 1, with 1,000 draws. Each draw resamples the announcements, re-estimates the daily model on resampled 66-day blocks, and rebuilds and re-estimates the monthly data with randomly sign-flipped errors. r^* and the FCI weights are held fixed.